Eric Romero
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Homework 11

Algorithm Description:

I proceed by seeing intervals as split moments, with start and finish times. Intervals have nonzero lengths, where it is critical that for every interval, it has to start before it ends, meaning for an interval i, there should be an edge between i_{start} to i_{finish}. This graph will become a recorder of all timings without knowing times.

Cases:

a) X ends before Y starts
   i) Edge from X_{finish} to Y_{start}
b) X and Y overlap
   i) Edge from X_{start} to Y_{Finish}
   ii) Edge from Y_{start} to X_{Finish}

Once the graph is constructed, run a topological sort.

If sort fails, then that means there is a directed cycle. This would mean that there is a chain of events that happen before itself.

If it succeeds, then it is considered legal. Using this ordering, we label each node its position within the topological order.

Correctness:

We need to show that legal start and end times are returned when statements are logically consistent.

We build the graph where every directed edge from nodes a to b means that event a must happen before event b. Due to the topological sort, every edge points forward in lieu of that ordering found from the sort. This means that the assigned time of a should be less than the time of b.

i. Suppose the algorithm returns start and finish times. This confirms the graph has a legal ordering from the topological ordering. For every interval X, an edge is created from X_{start} to X_{finish}, where timings means $X_{start} < X_{finish}$. This confirms intervals have a non-zero length.

Let's consider two intervals X and Y.

Case: Interval X ends before Y starts.

An edge is added for X_{finish} to Y_{start}. Because the edge points forward due to the ordering, assigned time for X_{finish} is earlier than Y_{start}.

Case: Interval X and Y overlap

Two edges are added from X_{start} to Y_{finish} and Y_{start} to X_{finish}. This confirms the relationship of two intervals overlapping one another.

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Both of these cases are satisfied by the times that are returned by the algorithm.

ii. Suppose the algorithm is unsuccessful in that statements are inconsistent. This means that there is a cyclical ordering found within the graph. As we have come to see, a directed cycle is when there are n nodes, and as the ordering moves forward from node 1 to 2 … to k, k will point back to node 1. This implies node 1 happens before itself, which is improbable, a contradiction that there is a consistent assignment of start and end times.

Time & Space Complexity

Let n: Number of intervals

Let m: Number of statements

Consider an interval, the graph will include two nodes representing its start and end times -> total vertices (|V|) = 2n.

For every interval, an edge is added from its own start node to its finish node -> n edges.

For the case where two intervals overlap -> two edges are added: one edge from one interval start to the other finish per interval.

In the worst case, m statements add 2 edges

Maximum possible edges: $|E| <= n + 2m$

This should mean the worst case time complexity: $O(n + m)$ for 2n vertices and adding at most n + 2m edges.

Topological Sort: $O(|V| + |E|)$

Worst Case Time complexity: $O(2n + n + 2m) \rightarrow O(n + m)$

Total Space: $O(n + m)$